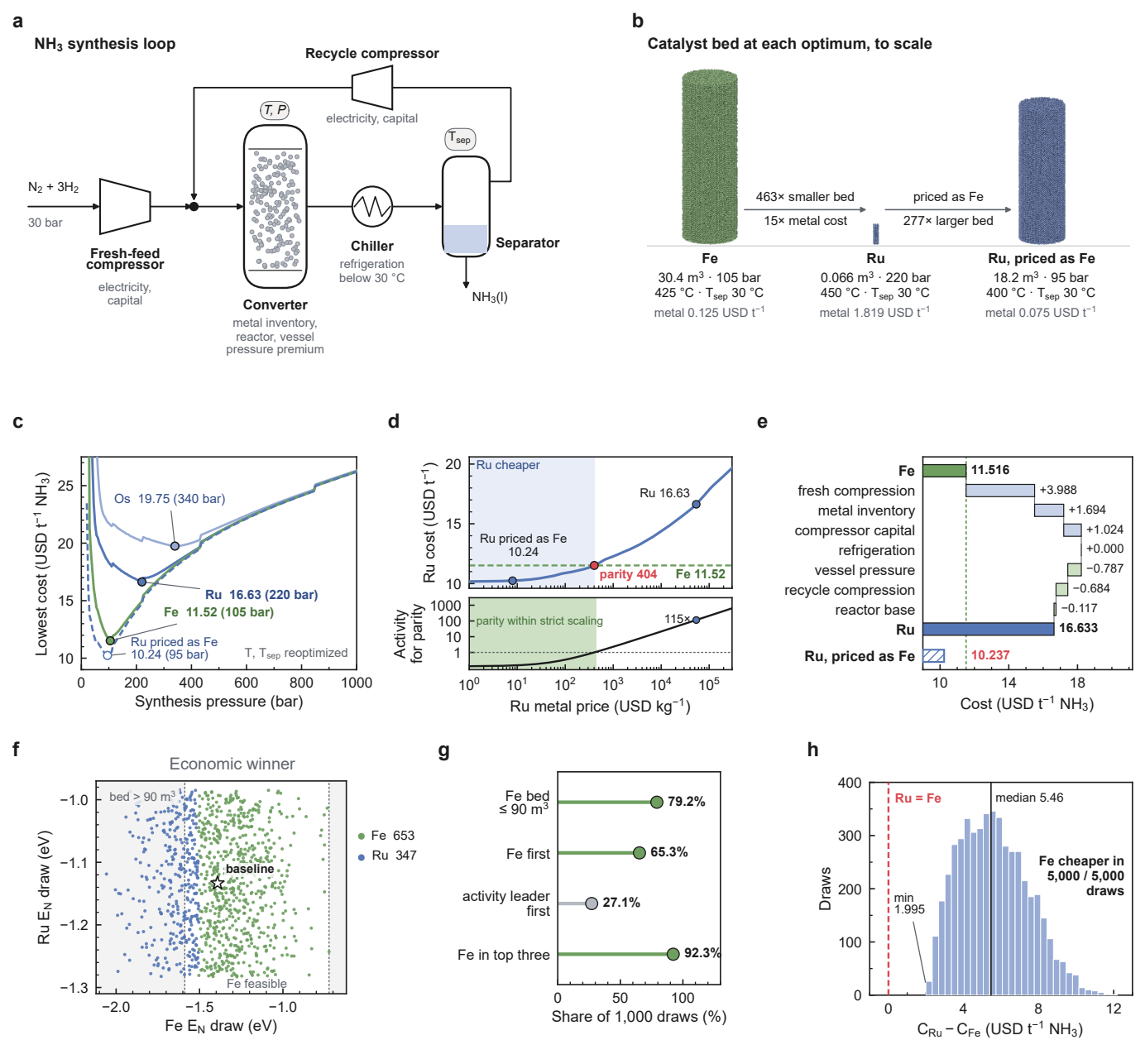

a
NH3 synthesis loop
Recycle compressor
electricity, capital
T, P
Tsep
N2 + 3H2
30 bar
Fresh-feed compressor
electricity, capital
Converter
metal inventory, reactor, vessel pressure premium
Chiller
refrigeration below 30 °C
Separator
NH3(l)
b
Catalyst bed at each optimum, to scale
463× smaller bed
15× metal cost
priced as Fe
277× larger bed
Fe
30.4 m3 · 105 bar
425 °C · Tsep 30 °C
metal 0.125 USD t−1
Ru
0.066 m3 · 220 bar
450 °C · Tsep 30 °C
metal 1.819 USD t−1
Ru, priced as Fe
18.2 m3 · 95 bar
400 °C · Tsep 30 °C
metal 0.075 USD t−1
c
Lowest cost (USD t−1 NH3)
Os 19.75 (340 bar)
Ru 16.63 (220 bar)
Fe 11.52 (105 bar)
Ru priced as Fe 10.24 (95 bar)
T, Tsep reoptimized
Synthesis pressure (bar)
d
Ru cheaper
Ru 16.63
Ru priced as Fe 10.24
parity 404
Fe 11.52
Ru cost (USD t−1)
parity within strict scaling
115×
Activity for parity
Ru metal price (USD kg−1)
e
Fe 11.516
fresh compression +3.988
metal inventory +1.694
compressor capital +1.024
refrigeration +0.000
vessel pressure −0.787
recycle compression −0.684
reactor base −0.117
Ru 16.633
Ru, priced as Fe 10.237
Cost (USD t−1 NH3)
f
Economic winner
bed > 90 m3
baseline
Fe feasible
Fe 653
Ru 347
Ru EN draw (eV)
Fe EN draw (eV)
g
Fe bed ≤ 90 m3 79.2%
Fe first 65.3%
activity leader first 27.1%
Fe in top three 92.3%
Share of 1,000 draws (%)
h
Ru = Fe
median 5.46
Fe cheaper in 5,000 / 5,000 draws
min 1.995
Draws
CRu − CFe (USD t−1 NH3)